Customer Case Intelligence - Project Report

Customer Case Intelligence | Fictional demonstration

Customer Case Intelligence

Return & Refund Intelligence. One case. Every connection.

Scope and source of truth

A specialist asks where a device return or refund stopped, what the bank has acknowledged, and which team needs which evidence. This is a fictional case investigation application, not a live telecom or bank system.

The corpus contains 87 synthetic source records across eight cases plus four attributed public reference notes, distributed as 91 PDFs. Public PDFs are authored paraphrases with original links; they are not official bank records or copies of bank contracts. All customers, identifiers, amounts, statuses, conversations and bank acknowledgements are fictional. Source facts in data/documents.json drive deterministic case reconciliation; the searchable narrative is extracted from the PDF corpus.

Ingestion, cleaning and freshness

The reproducible authoring records become PDFs under data/corpus; manifest.json records source identity, role, path and SHA-256. The loader verifies hashes, extracts page text, removes running headers and footers, normalizes whitespace, and reconstructs tables with headers before chunking. This demonstration is a versioned October 17, 2026 snapshot; ingestion runs on rebuild, with no scheduled refresh or live freshness SLA. Production requires provider-specific events, source timestamps, deletion handling and a negotiated freshness SLA.

Chunking and embedding

Fixed chunking uses LangChain recursive token splitting with a 350-token limit and 40-token overlap. Semantic chunking embeds adjacent sentences/paragraphs, splits at cosine-distance boundaries above the 70th percentile, and enforces the same token cap; the table-aware application strategy groups sections and repeats table headers when splitting rows. Each chunk includes source ID, case ID, role, PDF page, source hash, strategy and chunk ID. BGE small English v1.5 produces 384-dimensional embeddings; this local English model and short chunks are a measurable project baseline, not a claim of universal optimality.

Retrieval, reranking and storage

Authorized case and document-role filtering happens before dense or keyword ranking. BM25 and dense vectors each contribute their top 20 candidates to reciprocal rank fusion; a shared top-20 pool is optionally reranked with the trained MiniLM cross-encoder. The app defaults to fixed-size hybrid retrieval, with an explicit per-question cross-encoder option because reranking improves aggregate recall here but adds substantial latency. The app requests ten passages, groups them into at most six sources, and grounds responses on those excerpts. Cached chunk text and metadata live in ignored data/index/*.chunks.json; embedding vectors live in data/index/*.npy and are normalized in memory for cosine similarity. BM25 is built in memory over authorized candidates. PDF extraction caches live in ignored data/index/extracted/; model downloads live under ignored work/. PDFs and the provenance manifest are versioned; local indexes, identities, sessions, histories and keys are not published.

Stateful investigation and refusal

LangGraph enforces scope, retrieval, deterministic financial reconciliation, generation and citation verification. Foreign case identifiers, unknown identifiers, absent supporting evidence, unavailable providers and invalid exact

quotes have explicit refusal or fallback paths. OpenAI is optional and reads its key from the private environment; without it the app returns labeled excerpts. Citation checks validate exact quotes, not the semantic truth of every generated interpretation.

Access control

Users are authenticated with scrypt password hashes and opaque, hashed, revocable eight-hour server sessions. Passwords are created locally with the administration script, not committed or placed in reports. Every role also needs explicit case assignments; head access is organization-scoped through those assignments.

Role	Permitted source classes	Restricted examples
Customer service	Order, return, shipping, conversation, approved shared bank summary	Payment, approval, processor record, private statement
Supervisor	Service sources plus payment, routing, inspection, approval, refund	Private statements and account history
Head of operations	All assigned case sources	Other unassigned cases

APIs, retrieval, PDF downloads, handoff exports and saved runs enforce server-side access. Saved runs are isolated by username and current role. UI role labels do not grant permissions. Lower roles reconcile only their visible sources, so missing restricted evidence must not be presented as a universal negative. Customer service finance detail is explicitly restricted. Production still needs federated identity, organization tenancy, audit retention, encrypted storage, backup and security assessment.

Connected return and bank lifecycle

The order, return authorization, shipment, warehouse inspection, refund approval, processor submission, bank acknowledgement and statement have distinct sources. Shared bank summaries connect the case/order/bank-case identifiers, state, required information and next responsible team. Merchant submission does not prove bank receipt; bank receipt does not prove posting; provisional dispute credit is separate from final merchant credit. Missing bank information remains unknown. Suggested ownership is a human-review draft, not an external assignment.

Optional bank integration

app/integrations/banking.py provides an optional read-only Plaid sandbox transaction-sync adapter with pagination and sanitized failures. It is not wired to live customer accounts or automatically used to adjudicate refunds. It requires private sandbox credentials and access tokens. No refunds, disputes, bank transfers, customer messages or external tickets are executed. Transaction feeds alone do not prove universal bank acceptance; explicit provider references, account authorization and review are required. A documentation MCP can assist developers but cannot bypass bank consent or account access.

Official integration references: https://plaid.com/docs/api/products/transactions/ and https://plaid.com/docs/api/sandbox/ . TrueLayer's refund-return documentation illustrates that provider events may change and need verified signatures: https://docs.truelayer.com/docs/payout-and-refund-returns and https://docs.truelayer.com/docs/payments-api-webhook-reference .

Evaluation results for the PDF corpus

The final comparison uses the same 32 development questions on 91 PDFs (202 extracted pages), with frozen source-ID evidence spans and whitespace normalization for PDF reflow. The tokenizer is explicitly cl100k_base in every configuration. Fixed-size hybrid retrieval achieved 87.5% evidence recall@6, increasing to 89.1% with cross-encoder reranking; semantic retrieval achieved 82.8%, increasing to 85.9%. Fixed reranking improved recall

by 1.5625 percentage points and MRR by 0.060; semantic reranking improved recall by 3.125 points and MRR by 0.017. The table-aware configuration with reranking scored 84.4% on the primary benchmark. All eight supplemental financial-table questions achieved full evidence recall in all four tested table configurations.

Chunk counts were 274 fixed, 989 semantic and 342 table-aware. Warm median retrieval latency was approximately 35-37 ms without reranking, 7.86 seconds for fixed reranking, 2.52 seconds for semantic reranking and 5.14 seconds for table-aware reranking. These timings include concurrent local workload and are not isolated production benchmarks. The app defaults to the fast fixed-size baseline and lets users request deeper reranking per question; that choice exchanges latency for the measured aggregate recall improvement.

Every condition used the same questions, BGE embeddings, top-20 fusion pool and six returned passages. The report records per-query ranks, source IDs and corpus/query hashes. Earlier evaluation-expanded reports are historical pre-PDF results; preliminary PDF measurements before tokenizer correction are superseded. OpenAI generation faithfulness remains unmeasured without credentials. Synthetic templates and a small development question set limit generalization.

Customer Case Intelligence — Return & Refund Intelligence

One case. Every connection.

Assignment framework

My RAG app helps support and escalation specialists determine where a device return or refund stalled, using case documents and public reference guidance in an investigation workspace, with a target of 95% claim faithfulness and a warm-query latency target below 10 seconds. Targets are goals, not measured results.

Field	Decision
Use case	Support and escalation specialists ask where a device return/refund stalled, what evidence supports the finding, and what is needed next, in a web investigation workspace.
Corpus	The English corpus contains 87 synthetic order, shipping, inspection, payment, statement, support and shared bank records across eight fictional cases, plus four attributed reference notes, stored as 91 PDFs. Source-of-truth ownership remains with each originating system; public guidance never proves an individual case event.
Ingestion + cleaning	Reproducibly authored PDFs are hash-verified and read page by page; running headers/footers and excess whitespace are removed while tables, identifiers, amounts, and dates are retained. Every chunk carries case, source, document ID, source type, and revision metadata.
Ingestion + freshness	Initial ingestion occurs at startup from the versioned local snapshot; changed files require a service restart and index rebuild. This demo has no live institutional integrations or live freshness SLA; public references are dated and must be reviewed before real operational use.
Chunking + embedding	The main path preserves document sections and table headers, targeting 350 tokens with a tokenizer-enforced cap for ordinary prose and table blocks, using local BAAI/bge-small-en-v1.5 embeddings; a fixed-size baseline uses the same corpus and embedding model. Embedding-distance semantic segmentation is compared separately in evaluation rather than treating section boundaries as semantic chunking.

Field	Decision
Retrieve	A local persisted NumPy vector matrix plus BM25 forms a hybrid index; reciprocal rank fusion combines the top 20 candidates from each retriever, then a trained MiniLM cross-encoder reranks the fused top-20 shortlist and returns six passages. Case filtering is applied before ranking, and missing case evidence triggers abstention rather than borrowing another customer's records.

Scope and integrity

- All customer records, amounts, device IDs, timelines, and teams are fictional; no institution connection is implied.
- Refund approval, processor submission, bank posting, and provisional dispute credit remain distinct.
- No payments, external tickets, customer messages, or dispute decisions are executed.
- Escalation packets are downloadable drafts for human review.
- OpenAI answers require a server-side key. The no-key mode is evidence search/extraction, explicitly labeled.
- Server-enforced role and case permissions are implemented. Production still needs SSO, organization tenancy, audit retention, approved connectors and institution-specific policy review.

Chunking comparison and cross-encoder impact

PDF corpus: same 32 primary queries; frozen source-ID evidence spans, whitespace normalized for PDF reflow.

Controlled protocol

Explicit cl100k_base tokenizer. 350-token recursive baseline (40 overlap); semantic 70th-percentile adjacent embedding distance, same 350-token cap; BGE embeddings + BM25 RRF top-20 shortlist in every condition; return six chunks; no parent-document promotion during evaluation.

Trained reranker: Xenova/ms-marco-MiniLM-L-6-v2 (FastEmbed ONNX). Scores below are measured.

Configuration	Chunks	Evidence recall@6	Evidence precision@6	Evidence MRR@6	Document recall@6	Median ms
fixed	274	0.875	0.172	0.683	0.938	37.2
fixed_cross_encoder	274	0.891	0.172	0.743	0.938	7856.6
semantic	989	0.828	0.151	0.593	0.922	34.8
semantic_cross_encoder	989	0.859	0.161	0.610	0.922	2522.2
structure_cross_encoder	342	0.844	0.161	0.716	0.938	5138.6

Paired reranking impact

- fixed: evidence recall +1.56 percentage points; precision +0.00 points; MRR +0.060; median latency +7819.3 ms.
- semantic: evidence recall +3.12 percentage points; precision +1.04 points; MRR +0.017; median latency +2487.4 ms.

Failure analysis

fixed: 5 incomplete-evidence questions

- CASE-1008: Is the dispute credit provisional or a merchant refund? (evidence recall 0.00)
- CASE-1002: Was the money sent and is receipt visible to the customer? (evidence recall 0.00)
- CASE-1003: Connect the uninspected second device with the partial credit. (evidence recall 0.00)
- CASE-1004: Connect the outgoing refund and the incoming credit. (evidence recall 0.50)
- CASE-1008: Distinguish the merchant payment record from the temporary dispute adjustment. (evidence recall 0.50)

fixed_cross_encoder: 6 incomplete-evidence questions

- CASE-1002: Was the money sent and is receipt visible to the customer? (evidence recall 0.00)
- CASE-1003: Connect the uninspected second device with the partial credit. (evidence recall 0.50)
- CASE-1004: Connect the outgoing refund and the incoming credit. (evidence recall 0.50)
- CASE-1006: Compare the purchased device identifier with the scanned return. (evidence recall 0.50)
- CASE-1007: Does internal approval mean the customer received a credit? (evidence recall 0.50)
- CASE-1008: Distinguish the merchant payment record from the temporary dispute adjustment. (evidence recall 0.50)

semantic: 6 incomplete-evidence questions

- CASE-1002: Was the money sent and is receipt visible to the customer? (evidence recall 0.00)
- CASE-1003: Connect the uninspected second device with the partial credit. (evidence recall 0.00)
- CASE-1004: Connect the outgoing refund and the incoming credit. (evidence recall 0.00)
- CASE-1005: Does generating return paperwork establish that the carrier took possession? (evidence recall 0.00)
- CASE-1007: Does internal approval mean the customer received a credit? (evidence recall 0.50)
- CASE-1008: Distinguish the merchant payment record from the temporary dispute adjustment. (evidence recall 0.00)

semantic_cross_encoder: 7 incomplete-evidence questions

- CASE-1002: Was the money sent and is receipt visible to the customer? (evidence recall 0.00)
- CASE-1003: Connect the uninspected second device with the partial credit. (evidence recall 0.50)
- CASE-1004: Connect the outgoing refund and the incoming credit. (evidence recall 0.00)
- CASE-1005: Does generating return paperwork establish that the carrier took possession? (evidence recall 0.50)
- CASE-1006: Compare the purchased device identifier with the scanned return. (evidence recall 0.50)
- CASE-1007: Does internal approval mean the customer received a credit? (evidence recall 0.50)
- CASE-1008: Distinguish the merchant payment record from the temporary dispute adjustment. (evidence recall 0.50)

structure_cross_encoder: 6 incomplete-evidence questions

- CASE-1008: Is the dispute credit provisional or a merchant refund? (evidence recall 0.00)
- CASE-1002: Was the money sent and is receipt visible to the customer? (evidence recall 0.00)
- CASE-1003: Connect the uninspected second device with the partial credit. (evidence recall 0.50)
- CASE-1004: Connect the outgoing refund and the incoming credit. (evidence recall 0.00)
- CASE-1006: Compare the purchased device identifier with the scanned return. (evidence recall 0.50)
- CASE-1008: Distinguish the merchant payment record from the temporary dispute adjustment. (evidence recall 0.00)

Limits

- Eight fictional cases; template-assisted synthetic text is not representative of real institutional data.
- The 32 questions are a development benchmark, not an independently held-out test.
- Evidence annotations identify selected sufficient passages; unannotated supporting paraphrases can be scored as irrelevant.
- This report uses passage-level top-six metrics; the prior short-corpus report used unique-document metrics and is not directly comparable.
- Cold model downloads, index building, and OpenAI generation are excluded from warm retrieval timings.

Faithfulness: Not measured without live generation and a judge. Exact evidence-span recall and quote existence are not faithfulness.

Per-query ranks, source IDs, chunk IDs, model scores, and corpus/query hashes are recorded in evaluation.json. No query or gold annotation is loaded by the application retriever.

Interpretation

The cross-encoder improves aggregate evidence recall for both fixed and semantic chunking in this PDF experiment, at substantial latency cost. The app offers fast fixed retrieval by default with optional deeper reranking. Timings include concurrent local validation, and exclude cold downloads/index building and OpenAI generation.

Supplemental financial table evaluation

Eight supplemental financial allocation table questions; not included in the primary 32-query result. Requires both the tax amount and the control total from the designated order document.

Configuration	Evidence recall@6
fixed	1.000
fixed_cross_encoder	1.000
structure	1.000
structure_cross_encoder	1.000

This conservative source-specific probe can expose confusion between similarly worded allocation tables in order, payment, approval, and refund documents. It measures evidence retrieval, not generated arithmetic accuracy.

Implementation provenance

This repository is an original implementation of the customer device return and refund investigation use case. The academy solution kit is a requirements reference only: no kit code, synthetic records, prompts, reports or assets are copied into this repository.

The implementation uses public framework interfaces from LangChain, LangGraph, FastAPI, FastEmbed, ReportLab and PyMuPDF. The local source PDFs are generated from original fictional records; the public guidance notes are attributed paraphrases with original URLs. Optional bank integration follows the official Plaid sandbox transaction API; it does not access real accounts.

The comparison reports measure retrieval on declared synthetic development questions. No claims of production deployment, real bank acceptance, or measured OpenAI faithfulness are made without corresponding evidence.

Validation of the revised PDF application

The final full test run passed 44 tests. Following the extraction-cache relocation, 16 focused API/security tests passed; 22 focused financial, session-policy and bank-adapter tests passed, including two additional bank-lifecycle invariants. The 46 distinct tests are not claimed to have run as one final combined command.

A live local server was checked with all three privately generated roles: case visibility, PDF exports and denial of restricted statement URLs passed. The browser sign-in layout was visually reviewed. JavaScript syntax, Python compilation, Ruff formatting and undefined-name/import checks passed.

All 91 corpus PDFs were readable and their manifest hashes verified. Representative long documents, financial tables and bank summaries were visually reviewed. The final fixed strategy produced 274 unique chunk IDs, with the explicit cl100k_base token cap checked. Supplemental table retrieval found all required source-specific tax/control-total evidence for all eight cases in all four conditions.

The primary comparison is docs/evaluation-pdf.json and .md, using the final corrected tokenizer. Historical pre-PDF reports and preliminary measurements must not be substituted for that version. OpenAI generation quality is unmeasured; bank API tests use mock transports. No real payments or external customer actions were executed.

The publication scanner checks allowlisted source and extracted PDF text against common credential patterns and privately configured secret values without printing those values. Runtime files, credentials, SQLite identities/sessions/history, extracted caches, vector indexes and model weights are excluded. This is a useful publication check, not a guarantee that every possible secret format is detectable.